AMERICAN JOURNAL OF INDUSTRIAL MEDICINE 48:182–193 (2005)

Job Strain and Autonomic Indices of Cardiovascular Disease Risk

Sean M. Collins, PT, ScD, CCS,[1]* Robert A. Karasek, PhD,[2] and Kevin Costas, MPH[2]

Background *Despite the epidemiological evidence linking job strain to cardiovascular disease, more insight is needed into the etiologic mechanisms. This, in turn, would help to more precisely identify risk.*
Methods *We measured Job Strain using the Job Content Questionnaire, 8/day diary reports, and nationally standardized occupational code linkage, as well as autonomic regulation utilizing heart rate variability including spectral-derived components and QT interval variability in 36 healthy mid-aged males with varying strain jobs. The subjects wore Holter-monitors for 48 hr; this included a work and rest day.*
Results *Job strain ($P = 0.02$) and low decision latitude ($P = 0.004$) were associated with a reduction in cardiac vagal control (HFP) persisting throughout the 48 hr. Job strain was also associated with elevations in sympathetic control during working hours ($P = 0.003$).*
Conclusions *The disturbed cardiovascular regulatory pattern associated with job strain may help explain the increased risk of cardiovascular diseases linked with occupational exposure.* Am. J. Ind. Med. 48:182–193, 2005. © *2005 Wiley-Liss, Inc.*

KEY WORDS: job strain; heart rate variability; job content questionnaire; psychosocial factors; occupational stress; cardiac disease risk

INTRODUCTION

Substantial research now exists relating adverse psychosocial work environments to cardiovascular disease, with a recent comprehensive review of 46 studies concluding: ''the strong, consistent evidence of an association . . .support the conclusion that job strain is indeed a major CVD risk factor'' [Schnall et al., 1998; Belkic et al., 2004]. The public policy implications of confirming ''work-relatedness'' are significant since cardiovascular disease remains the leading cause of mortality and morbidity in industrialized countries. Indeed, the recent Korean practice [KOSHA, 1998] of classifying significant portions of cardiovascular illness as work-related has elevated heart disease to one of the two primary causes of work-related mortality in Korea. Gaps remain in this research, however, notably: there is less research on the specific physiological pathways of disease causation [Schnall et al., 1994, 2000], although failures in blood pressure regulation, when assessed by ambulatory monitoring, are clearly identified [Schnall et al., 1998]. We assess whether regulatory limitations of nervous system control (sympathetic and parasympathetic autonomic) could be one of the important pathways by which psychosocial risks contribute to cardiovascular illness. These limitations, if confirmed, could implicate work stress in the broad range of other diseases affected by autonomic nervous system (ANS) regulation.

The demand control model (or demand/control/support model) is perhaps the most commonly utilized hypothesis associating psychosocial factors at work with cardiovascular

[1]Department of Physical Therapy, School of Health and Environment, University of Massachusetts, Lowell, Massachusetts
[2]Department of Work Environment, School of Health and Environment, University of Massachusetts, Lowell, Massachusetts
Institution where work was performed: Department of Work Environment, School of Health and Environment, University of Massachusetts Lowell.
Contract grant sponsor: Chancellor's Research Fund; Contract grant sponsor: Kerr Ergonomics Institute; Contract grant sponsor: Department of Work Environment, School of Health and Environment, University of Massachusetts Lowell.
*Correspondence to: Sean M. Collins, School of Health and Environment, University of Massachusetts Lowell, 3 Solomont Way, Suite 5, Lowell, MA 01854-5124.
E-mail: Sean_Collins@uml.edu

Accepted 24 June 2005
DOI 10.1002/ajim.20204. Published online in Wiley InterScience (www.interscience.wiley.com)

© 2005 Wiley-Liss, Inc.

disease [Karasek, 1976, 1979; Johnson and Hall, 1988; Karasek and Theorell, 1990]. The demand—control model classifies jobs based on psychological demands (such as work deadline pressure) and decision latitude (control over the work situation). The primary hypothesis involves high psychological demands coupled with low decision latitude (defined as ''job strain'') increasing the risk of cardiovascular disease [Schnall et al., 1994, 2000]. However, a relative paucity of strong associations between job strain and conventional risk factors, other than blood pressure, suggests that the job strain coronary heart disease association is mediated through different physiological pathways. Sudden cardiac death (SCD)—which accounts for approximately 50% of cardiac mortality—results from defective control of cardiac rhythms [Babuty and Lab, 2001; Gronefeld and Hohnloser, 2001; Lombardi et al., 2001]. The common opinion is that SCD occurs primarily in individuals with conventional risk factors, however, these factors are not always the precipitating cause of SCD [Myerburg and Spooner, 2001]. Defective cardiac regulation is one pathway to cardiovascular disease implicated by job strain research which can be simply and non-invasively monitored with a Holter electrocardiagram monitor.

The central nervous system's autonomic branches influence the heart's electrical activity through the sympathetic (SNS) and parasympathetic (PNS) divisions of the ANS, influencing heart rate and beat-to-beat rhythm. Parasympathetic activity promotes relaxation and regeneration and sympathetic activity mobilizes the body for action such as the ''fight or flight'' response [Cannon, 1914; Guyton and Hall, 2000]. Heart rate variability (HRV) provides an assessment of the ANS as it influences heart rate [Bigger, 1994; Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996; Berntson et al., 1997].

Low HRV, predicated on its ability to assess ANS based cardiac regulation, has been utilized in clinical studies of CVD patients to predict a variety of cardiovascular outcomes [Kristal-Boneh et al., 1995; Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996; Fetsch et al., 1998; Stein and Kleiger, 1999], and in healthy populations, all cause and cardiac mortality [Algra, 1993; Tusji and Vendetti, 1994; Tusji and Larson, 1996; Dekker, 1997], as well as new onset of hypertension [Singh et al., 1998]. Low high frequency power of the HRV spectrum is also a risk for mortality in patients who have had a myocardial infarction [Kleiger et al., 1987; Bigger, 1992; Lombardi et al., 1996; Bigger et al., 1998; Doulalas et al., 2001], have coronary artery disease [van Boven et al., 1998] or congestive heart failure [Mortara and LaRovere, 1994; Nolan, 1998; Galinier et al., 2000].

There is a broader public health implication here also: stress theories are non-specific in predicting chronic disease from social stressors. If the implicated pathway is the central, basal regulatory power of the ANS, with its sympathetic and parasympathetic mechanisms, this might impact upon a broad range of the chronic diseases potentially associated with job strain [Van der Doef and Maes, 1999; Bongers et al., 2002].

Studies utilizing HRV measures for investigating the impact of occupational stress in the workplace exist, but with limitations on recruitment of subjects and exposure assessment strengths [Collins, 2001]. Most job strain etiologic studies above have relied on assessment methods such as questionnaires (Job Content Questionnaire (JCQ) and ERI) [Karasek et al., 1998], a smaller number have used occupational linkage methods. The use of multiple types of job strain assessments, the triangulation method of Kristenson [1995, 1996], can bolster validity of interpretation and is used in this paper.

MATERIALS AND METHODS

Study Population

A sample of healthy employed men ($n = 30$) between 35 and 59 years of age, all members of a community health plan, was recruited from an ongoing investigation of seasonal variations in cholesterol (Seasons Study) [Merriam et al., 1999]. An additional sample ($n = 6$) of healthy employed men (35–59 years of age) was recruited from an associated stress reduction clinic. Participants of the Seasons study had been screened for job strain using a subset of questions from the Job Content Questionnaire [Karasek et al., 1998]. Investigators reviewed the JCQ and current occupational information collected for each subject as part of the intake process and identified a potential sub-population of and high and low strain individuals. Potential subjects were sent letters explaining the study, how they were selected and that they would be contacted by telephone. The Seasons population excluded individuals with blood lipid morbidity or related therapies, psychiatric illness, known likely-to-be-fatal diseases, and study participation limiting conditions (such as alcoholism). Our recruitment protocol further excluded subjects with serious medical conditions leading to medications that would affect HRV monitoring, or heart disease, or any medications affecting HRV monitoring. The subject population is similar to the rather homogenous Seasons recruiting base population in gender, age, breadth of occupations (see below), white ethnicity, and in social class. Only day-shift workers were recruited, again similar to the Seasons population. Subject recruitment occurred in two waves. During the first wave of recruitment, difficulties associated with participation in complex protocol studies for participants with high strain jobs became evident. The yield of subjects screened as having a probable high strain occupation (17%) was much less than that for subjects screened as

having a probable low strain occupation (42%), potentially biasing results. The recruitment protocol was made more flexible to accommodate the high strain individuals without influencing critical elements of the protocol's scientific method and recruitment of subjects screened as being probable of being in a high strain occupation improved (83%). Holter monitor attachment occurred at a location and time of the subject's convenience, increased monetary incentives were offered (up to $350), and the work rest day cycle could be any two days of their regular workweek. A total of 36 subjects were successfully recruited and monitored.

Subjects are from a population based sample and, therefore, comprise a wide range of occupations from a range of employment sectors. Subjects in high strain occupations held positions such as letter carrier, laborer, printer, quality technician, and tele/data communications; while subjects in low strain jobs held positions such as teacher, business agent, retirement specialist, and buyer. We feel this diversity is a significant strength of this sample as opposed to the limited occupational variation when collecting data from one sector or organization.

Protocol

The Fallon Community Health Plan and the University of Massachusetts Medical Center Institutional Review Board approved the study and subjects provided informed consent prior to participation. Subjects completed a full version of the Job Content Questionnaire and provided occupation and demographic information. They were connected to a Holter monitor (Marquette 8500) to collect a continuous 48 hr ECG recording beginning the morning of (prior to) a workday followed through a rest day.

INDEPENDENT VARIABLES

Job Strain

Kristensen's triangulation method places emphasis on multiple methodologically accurate measurements of risk factors [Phillips and Smith, 1993; Kristenson, 1996]. This paper utilizes a triangulation based on (A) self-reports to the Job Content Questionnaire and (B) workday diary, and (C) three-digit occupational linkages to US nationally representative job characteristic databases.

Subjects completed a diary with questions relating to decision latitude, psychological demands, activity, exertion, position, and social interaction at eight pre-determined times each of the monitored days (three pairs of observations separated by 20 min during work), which were aggregated over the work day for this classification. The diary was a modification of one utilized in a study of ambulatory blood pressure monitoring [Schnall et al., 1990]. Questions are similar in content to the JCQ, but modified to reflect short-term exposures. Diary scales with moderate reliability but good theoretical congruence with JCQ and linkage-based scales were constructed using factor analysis with varimax rotation. The aggregated self-report diary assessments provide a measure where the situation to be assessed is closer in time to the report, potentially reducing recall bias [Kristenson, 1995, 1996; Landsbergis and Theorell, 2000].

"Objective"—non-subject influenced assessment of job characteristics come from an occupation-based measure of occupational linkage. The occupational title provided by subjects' and their three digit occupational codes (i.e. doctor, carpenter. . .) consistent with the JCQ Center's coding of the US nationally representative Quality of Employment Surveys [Karasek et al., 1985] were recorded. The JCQ Job-Occupational linkage system provided national average psychological demands and decision latitude scores associated with each subjects' occupation code [Karasek et al., 1985]. A method to weight and combine the individual assessment scores was developed (Appendix I) transforming all scores into equivalent JCQ scale scores.

Classification of Strain, Demands, Control, Social Support, Socioeconomic Status

High strain jobs were classified using tertiles of the final scores based on the JCQ Center US national sample JCQ scale means and standard deviations. Subjects were combined in a nine cell bivariate matrix of decision latitude and psychological demands. Those in the most extreme three high demand and low control cells were categorized as high strain, yielding 13 high strain subjects. To test the univariate main effects of psychological demand, decision latitude, and social support on the dependent variables the weighted score for demand and control, and the original JCQ score for social support were utilized. National US JCQ sample means were used as cut-points to split the group into high ($n = 15$) and low ($n = 21$) demands as well as high ($n = 17$) and low ($n = 19$) control groups. Social support was added as a continuous scale variable to test for isostrain [Johnson and Hall, 1988].

DEPENDENT VARIABLES

Electrocardiograph Data Processing

Holter data tapes were analyzed for ventricular arrhythmias and HRV at the Research Holter Laboratory at Columbia University [Albrecht and Cohen, 1988; Rottman et al., 1990; Bigger, 1994; Sloan et al., 1994]. Dependent variables (high frequency power, low frequency/high frequency power ratio, HRV (measured with SDNN), and heart rate) were computed from 5-min epochs throughout the 48-hr period. A Fast Fourier algorithm computed power in the low frequency (LF, 0.04–0.15 Hz) and high frequency (HF,

0.15–0.40 Hz) spectra. High Frequency Power is an indicator of parasympathetic autonomic response [Malliani et al., 1991; Bigger, 1992, 1994; Berntson et al., 1997]. Three estimates of sympathetic activity are utilized and computed for each 5-min epoch. The first: the ratio of low to high power, has been often used, but is considered more ambiguous than the high frequency power with respect to parasympathetic response [Malliani et al., 1991; Pagani et al., 1991a,b; Berntson et al., 1997]. Residual heart rate, the second measure, has been proposed as a relative indicator of sympathetic activation by making use of the demonstrated relationship between vagal activity, the primary determinant of heart rate, and the heart rate itself. Positive residuals, for example, indicate that heart rate is higher than estimated by vagal activity, and sympathetic activity is proposed as the major source of this increase. Residual HR was determined by the methods proposed by Grossman [Grossman and Sveback, 1987] and further validated by Backs and Lenneman [Backs, 1995, 1998; Backs et al., 1999; Grossman and Sveback, 1987; Lenneman and Backs, 2000]. Weaknesses of the residual heart rate as an indicator of sympathetic measures are dependencies on the ability of HFP to assess parasympathetic activity and the assumption of linearity in the cardiovascular response to interacting sympathetic and parasympathetic activity [Backs, 1995, 1998; Backs et al., 1999].

QT variability index (QTVI), the third method, has been shown to be a marker of sympathetic nervous system activation [Yeragani et al., 2000], and to predict risk of sudden cardiac death [Berger et al., 1997; Atiga et al., 1998], and with certain emotional and mental health conditions that are known to be associated with elevated sympathetic nervous system activity such as panic disorder and depression [Yeragani et al., 2000]. QTVI measures the variability of the QT interval after controlling for the known effect of heart rate. Increased sympathetic nervous system activation leads to increases in QT interval variation that is independent of the HRV [Negoescu and Dinca-Panaitescu, 1997].

The digitized and edited ECG signals were submitted to algorithms developed at Johns Hopkins School of Medicine, Department of Cardiology. QT variability index (QTVI) was computed by the methods of Berger [Berger et al., 1997]. Long delays in the QT interval series resulting from noisy signals, artifact or ectopic beats that could not be filtered were eliminated from analysis (roughly 40%). The entire epoch was eliminated if more than 25% of the data could not be utilized.

Sociological Periods

Subjects recorded times on the diary indicating changes in their social situation, such as the start of work, end of work, etc. These times were utilized to construct ''sociological periods,'' such a ''morning at work,'' ''lunch,'' and ''afternoon at work'' (as seen in Figs. 1 and 2)—roughly seven periods per day. These periods allow control for diurnal rhythm variations that may be related to cardiovascular function. However, they allow for a more precise understanding of the social determinants of daily activity effects than arbitrary absolute time of day cut points often used in circadian research. The actual times of major socially mediated activities may be the most salient component of much diurnal variation [Schwartz and Pickering, 1996; Jacob et al., 1999]. The ECG data for each subject during each sociological period were aggregated for comparisons between groups within each sociological period.

Covariates

Covariates that have been shown to exert an influence on HRV controlled for in this study include age and gender. The recruitment protocol restricted the pool of potential subjects to men between the ages of 35 and 59 years of age. This moderate-cost HMO intake based population tends to suggest a mid-range socioeconomic position; however education was used to control for social class. Diary data reporting position and exertion were available at eight discrete times during each day, therefore, six of the 14 sociological periods have information available regarding position and exertion. Subjects were asked to fill out the diary based on the 20 min time period prior to completing the diary. Total cholesterol, body mass index, and smoking status were controlled in the 30 seasons study subjects.

Statistical Analysis

Graphic and descriptive analyses were performed to assess differences between high and low job strain groups during the sociological periods. To test for significant differences between the job strain, control and demand groups fixed effects, repeated measure models with the SPSS 11.1 mixed model procedure using an autoregressive covariance matrix were computed. Sociological period is the repeated measure with each subject uniquely identified within each period. Four models were constructed to test hypotheses for each dependent variable with $\alpha = 0.05$ as significant. All models included age and education. Additional parameters including social support and cardiovascular risk factors (BMI, total cholesterol, smoking) were tested without effect on the outcomes of the models. Model I tested the effect of Job Strain, entered as a dichotomous variable. Model II, tested the effect of job strain specifically when at work: an interaction variable was entered between location (0, not at work; 1, at work) and job strain. Models III and IV tested the main effects of decision latitude and psychological demands entered as dichotomous variables.

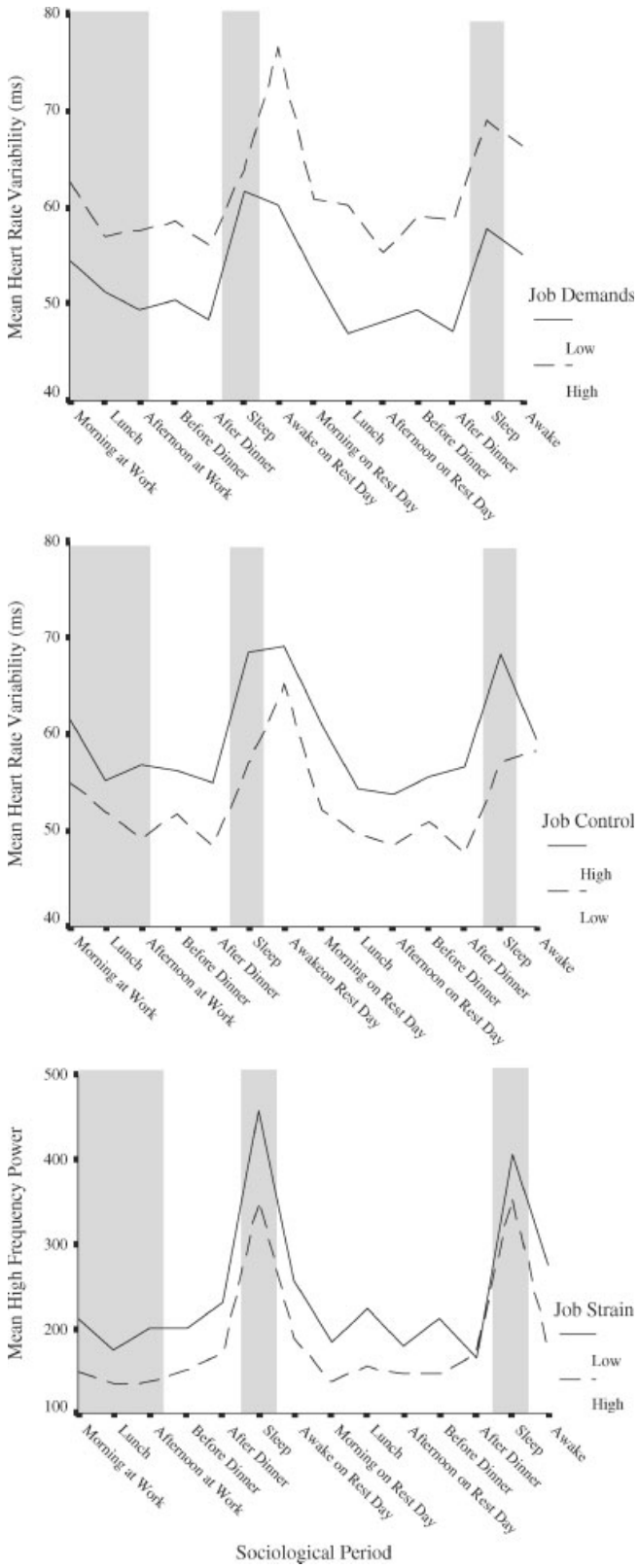


FIGURE 1. Heart rate variability and high frequency power during sociological periods.

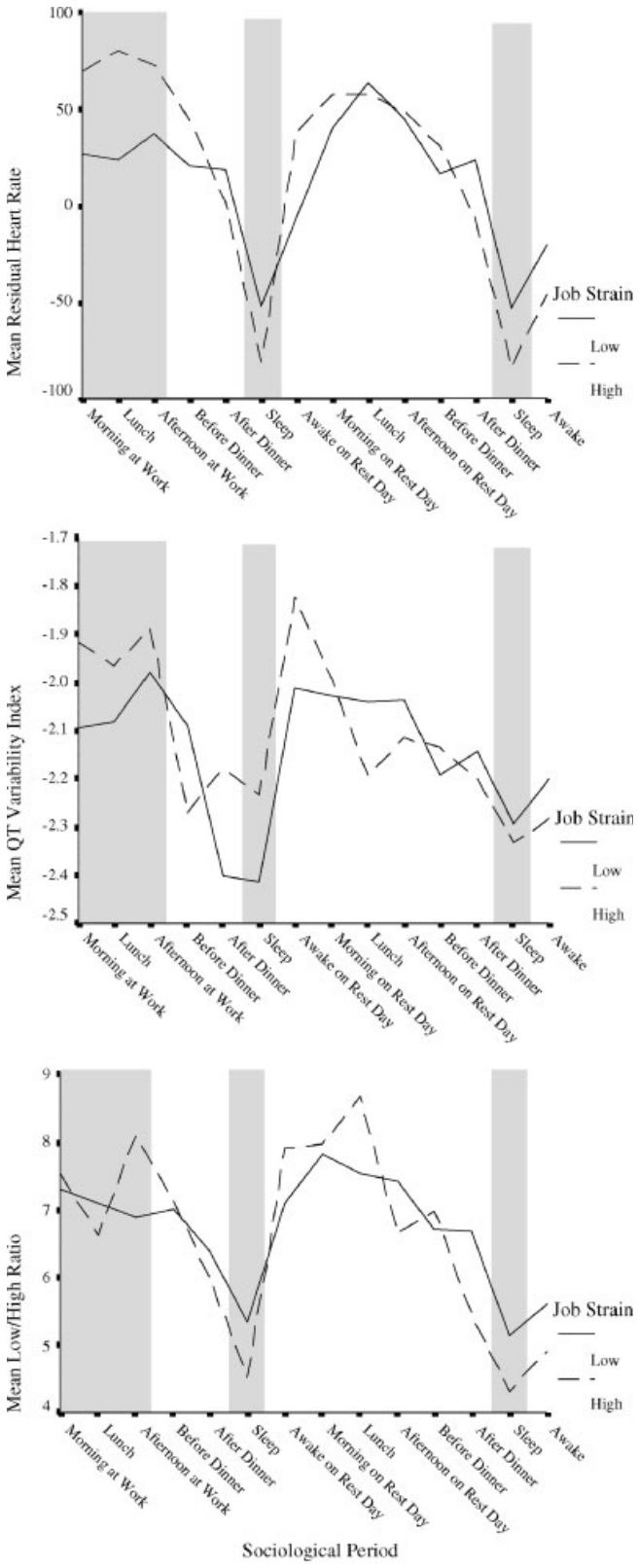


FIGURE 2. Sympathetic variables during sociological periods between strain groups.

RESULTS

The time variation of the ECG variables over the 48-hr period is presented in Figures 1 and 2 depict the differences between the strain groups within each sociological period. Parameter estimates and related statistics from the fixed effects repeated measures models for job strain, decision latitude, and psychological demands are presented in Tables I and II. Social support, smoking (all subjects), body mass index and total cholesterol (n = 30) were not significant predictors when added to any of these models.

Heart Rate Variability

Table I presents parameter estimates for HRV and HFP for four models. The models differ in the primary parameters (i.e. job strain, strain at work, demands, control) while the covariates (age and education) are included in all models. As presented in Table I there is no significant effect of high job strain on HRV either during the monitoring protocol (Model I) or while at work (Model II). Significant reductions in HRV were found in the low control group (Model III) and contrary to hypothesis (see Discussion), elevations were associated with increased demands (Model IV). Age was negatively associated with HRV in all models. Education was not associated with HRV in these models.

High Frequency Power—Parasympathetic Activity

Table I depicts model parameters for high frequency power. Significant differences exist between job strain groups (Model I) throughout the entire monitoring period. Increased age is associated with decreased high frequency power. The low job control group also demonstrated a significant reduction in high frequency power during the entire monitoring period (Model III), with increased education level also associated with decreased high frequency power. No such reduction was identified for psychological demands (Model IV), which has been a standard hypothesis in HRV research.

QTVI, Lo/Hi Ratio, and Residual Heart Rate—Sympathetic Activity

Table II presents model parameters for the sympathetic nervous system measures of QTVI, Lo/Hi Ratio, and the residual heart rate. All sympathetic measures displayed the same pattern of association with job strain. They were not significantly different between job strain groups when the model considered the entire monitoring period. However, all sympathetic measures were significantly elevated in the high strain group on the workday (Model II). Education was not significantly associated with sympathetic measures.

DISCUSSION

High job strain and, even more clearly, low decision latitude are associated with significant and persistent reductions in cardiac vagal control. Furthermore, at work, subjects with high job strain demonstrated elevations in sympathetic control as measured by three different ECG based parameters. While job strain does not predict total HRV, low decision latitude is associated with reduced HRV. These findings provide evidence of an association between job strain exposure and physiological processes known to be associated with cardiac deregulation. The persistent decrease in parasympathetic cardiac control (HFP) in the high strain

TABLE I. Model Parameters for Heart Rate Variability and High Frequency Power

		Heart rate variability			High frequency power		
	Parameter	Estimate	SE	Sig.	Estimate	SE	Sig.
Model I	High Strain	2.95	4.00	0.463	−111.97	46.73	0.019
	Education	0.932	1.04	0.373	−14.53	12.53	0.250
	Age	−0.612	0.278	0.032	−14.28	3.62	0.001
Model II	High Strain – Work	−2.77	4.02	0.491	−55.29	45.68	0.227
	Education	0.935	1.03	0.372	−14.56	12.52	.249
	Age	0.612	0.278	0.031	−14.21	3.61	0.001
Model III	Low Control	−8.26	4.18	0.050	−147.43	49.56	0.004
	Education	−0.528	1.13	0.645	−29.20	13.77	0.037
	Age	−0.545	0.147	0.001	−10.09	3.58	0.006
Model IV	High Demands	7.98	3.67	0.034	−74.09	45.38	0.112
	Education	0.291	0.967	0.765	34.20	23.27	0.099
	Age	−0.499	0.273	0.073	−12.53	4.32	0.003

Repeated measure mixed models procedure using an autoregressive covariance matrix (N = 34).

TABLE II. Model II Statistics for Sympathetic Variables

	Statistics for sympathetic variables			
Variable	**Parameter**	**Estimate**	**SE**	**Sig.**
		Model I		
QTVI	High strain	0.082	0.105	0.439
	Education	0.004	0.031	0.872
	Age	0.013	0.008	0.129
Lo/hi ratio	High strain	0.205	0.649	0.753
	Education	−0.113	0.209	0.513
	Age	0.139	0.051	0.008
Residual heart rate	High strain	7.46	8.28	0.370
	Education	0.423	2.17	0.812
	Age	0.507	0.639	0.430
		Model II		
QTVI	High strain-work	0.278	0.088	0.002
	Education	0.005	0.029	0.866
	Age	0.012	0.008	0.250
Lo/hi ratio	High strain-work	1.58	0.602	0.009
	Education	−0.099	0.179	0.581
	Age	0.142	0.057	0.008
Residual heart rate	High strain-work	68.23	11.69	0.001
	Education	0.415	2.07	0.841
	Age	0.495	0.599	0.410

Repeated measure mixed models procedure using an autoregressive covariance matrix (N = 34).

group and in the low control group and the persistent reductions in HRV in the low control group provide evidence for work relatedness of autonomic regulation of cardiac function known to be associated with elevated risk for several cardiovascular diseases. The protocol is based on the significant body of literature that has investigated and identified a relationship between job strain and cardiovascular disease while adding more sophisticated monitoring strategies [Berntson et al., 1997] and risk assessment methods [Bigger, 1992] based on well-defined parameters from the human factors and cardiovascular physiology and epidemiology literature. The ''triangulation'' job strain assessment method found substantial consistency between questionnaire and diary measures of job strain, with occupational codes adding further consistency for decision latitude.

Contrary to a decade of well-known laboratory research on mental workload [Kalsbeek, 1973; Backs, 1995; Kamarack et al., 1998] this study shows psychological demands to be positively associated with HRV. This may reflect a difference between the measurement situation of short acute stress exposures in the laboratory where demands reduce short-term variability versus 48 hr of variable exposure and readjustment in the context of daily social activity. The variability in long-term field exposures may reflect the body's arousal/return to baseline response to its normal daily sequence of unpredictable challenges [Conway et al., 1984]. This highlights the limitations of directly relating laboratory findings of physiological responses to field studies. Further consideration for the interpretation of HRV an indicator of demands in the field is required.

Some consistency is found with other recent job stress research. Vrijkotte et al. [2000] utilized Siegrist's model of effort reward imbalance [Siegrist et al., 1990] to assess vagal control in subjects reporting high and low effort-reward imbalance. Subjects with a high imbalance had significantly lower vagal cardiac control (time domain HRV parameter–RMSSD) and these effects were noted throughout the monitoring period on both the work and rest day.

Van Amelsvoort et al. [2000] tested the impact of job strain, noise, and shift work on HRV (including frequency domain measures). High strain subjects demonstrated elevated sympathetic arousal (as assessed by the percentage of Low Frequency HRV), however there were no significant differences between the strain classification groups in vagal cardiac activity (as measured by HFP). We have recruited subjects from a normal working population with a mean age of 45. Typical job tenures are measured in multiple years for such a group in US surveys (US Quality of Employment Survey), although we did not collect job tenure information specifically. Job mobility is much higher under age 30, which is outside our recruiting boundary. In contrast, van Amelsvoort et al. [2000] has specifically recruited newly employed subjects with a mean age of 30. This could reflect as much as an order of magnitude difference between the studies in job tenure. For this reason we find the results from van Amelsvoort et al. [2000] difficult to compare to our study. In addition to the subject characteristics, the manner in which subjects were divided into job strain exposure categories is quite different. Van Amelsvoort et al. [2000] divided subjects based on cut points from the sample mean. We feel it is necessary to cut study samples based on national means to minimize misclassification of jobs as high strain. In data analysis, they normalized data based on individual sleep values. They point out that this reduces between subject variance. We would point out that if changes to HRV last into the sleep hours (as we find with the High Frequency Power component of HRV) then this normalization substantially removes the between subject and between group variation that is associated with the exposure of interest.

Findings based on sympathetic assessments in this sample are strengthened by the consistency of responses for all three measures (QTVI, Lo/Hi ratio and Residual Heart Rate): significant differences are only identified for the workday. These might reflect, at least in part, the significant difference in exertion or posture between the strain groups on the workday. Posture might also explain the reductions in vagal control during the workday for high strain workers. However, other within subject analyses on this dataset [Collins et al., 2003] show that exertion does not differ by

strain groups and the difference in posture (high strain upright) does not account for a majority of the variance. Also, the persistence in reductions in high frequency power over the entire monitoring period leads us to believe that posture is not accounting for the differences in the groups in vagal control.

The job strain hypothesis suggests that environmental exposure to job strain can lead to exhaustion (such as vital exhaustion) that has been associated with altered vagal control [Appels and Mulder, 1988; Appels et al., 1988], and CHD [Prescott et al., 2003]. In testing the effect of vital exhaustion, Wantanabe et al. [2002] found significant reductions in HFP measured during 5 min of supine rest in subjects with a high vital exhaustion score as compared to low and moderate scores. Exhausted subjects may not be able to utilize the vagal system as a dominant control system for cardiac output regulation, leading to a sympathetic dominant control mechanism.

The observed stress responses are consistent with emerging understandings of etiologic pathways of disease development [Schnall et al., 1994, 2000]. Persistent decreases in vagal cardiac control may help explain elevated ambulatory blood pressure readings on work and rest days in subjects with high strain occupations. Such cardiac regulatory deregulation may be linked to increased risk of developing clinical hypertension, as well as development of pathological changes in the myocardium [Schnall et al., 1990]. Findings by Lucini et al. [2002] suggest that cardiac regulatory changes associated with reductions in vagal cardiac control, represent a spectrum of changes that may precede clinical hypertension. Important from a public health and primary medical care perspective is the fact that these cardiac deregulations occur prior to clinic-based evidence. If sudden cardiac death of an electrophysiological origin is related to alterations in vagal cardiac control, then one might hypothesize that pre-clinical cardiac deregulation can increase susceptibility to sudden cardiac death as hypothesized by Lombardi et al. [2001]. Cardiac disease related to alterations in regulatory control may not only be mediated through elevations in blood pressure, but can increase risk of sudden cardiac death.

While vagal cardiac control had prolonged reductions the indices of sympathetic control—QTVI, Lo/Hi ratio, and residual heart rate—are elevated in the high strain group on the workday. These findings may help explain elevated BP during the workday (due to increased sympathetic and decreased vagal activity) and then persistent elevations in BP after work (due to persistent drops in vagal control). This may also be responsible for the identification of increased cardiac reactivity in high stress subjects [Steptoe et al., 1996; Steptoe, 2000] even when tested after work.

The time component of causality reported in this paper is based on the fact that the 48 hr of cardiac control phenomena are all time-subsequent to the questionnaire administration at the beginning of the monitoring. More detailed within-subject analysis in for the job strain/HRV covariations are reported elsewhere (both between [JCQ, occupation] and within subject [diary] work/HRV associations remain significant). Furthermore, presumably the job strain risk has actually developed over a significantly long time period, but that full development process cannot be tested in this project. We would recommend this be tested in future research.

The independent variables shown to influence high frequency power between subjects in previous investigations have been age, sex, and physical fitness [Kristal-Boneh et al., 1995]. Study selection criteria assured that study subjects were male, between the ages of 35 and 59, generally healthy and not using medications known to affect HRV. Control for age increased the association between job strain and the HR variables, which often occurs in such field studies, and adding education also somewhat strengthened the association, which is less common. While the literature does not suggest a relationship between typical cardiovascular risk factors and atypical HRV [Kuch et al., 2000], the models were tested with control for cholesterol, body mass index and smoking status in 30 of the 36 subjects. These variables were not significantly associated with any of the tested outcomes. There is no evidence in the literature to suggest a relationship between decision latitude and physical fitness. It may be hypothesized that greater decision latitude is conducive to a lifestyle with greater potential for the pursuit of personal fitness. High decision latitude individuals might self-select toward better physical fitness, and this possible association should be considered in future investigations.

This study was performed among healthy men. Depressed HRV is well recognized as a powerful independent predictor of mortality and risk of life-threatening arrhythmias among patients after acute myocardial infarction (AMI). Return to high strain work is an independent predictor of lethal ischemic events among younger men post-AMI [Theorell et al., 1991]. The present study shows replicated evidence that high strain work adversely impacts upon indices of HRV. Examination of autonomic indices post-AMI in relations to return to work issues should also be given high priority in future studies.

Other work environment exposures might influence HRV measures through their influence on ANS function. Two possibilities include noise [van Amelsvoort et al., 2000; Tzaneva et al., 2001] and air particulate exposure [Magari et al., 2001, 2002], and these possible associations should be tested in future investigations.

More intervention studies are urgently needed which examine the impact of improving work conditions upon autonomic indices. One excellent example is Kobayashi et al. [1997], a study of the autonomic effects of modifying work hour patterns prior to engaging in night shift work. Other examples, could be developed following the job redesign

principles involving enhancement of task autonomy and job security related possibilities of maintaining social equilibrium as suggested in the demand/control model [Karasek and Theorell, 1990], and in the stress-disequilibrium model [Karasek, 2005].

Conducting this study required a significant effort to recruit a community-based sample with enough variation in job strain measures to provide the necessary contrasts to be assessed. During recruitment it became obvious to the researchers that high strain subjects were less likely to participate. To recruit high strain subjects, without sacrificing the essential aspects of the study protocol required, the researchers had to expend considerable effort to increase the flexibility of monitor hook up time, day, and location and provide additional monetary incentives. Analysis shows that without this major effort to recruit high strain subject pool using the remaining subjects to define relative high and low strain groups would have likely resulted in null findings. Future studies should extend to female subjects and minority populations.

CONCLUSIONS

This study supports the hypothesis that job strain is associated with ambulatory electrocardiograph indices of cardiovascular regulation in a pathogenic manner. The findings implicate autonomic deregulation under job strain conditions, providing a putatively powerful explanation of heart disease, as well a potential explanation of ''work-relatedness'' for other stress-related chronic diseases.

The emerging need to understand work organization risks has placed work-related psychosocial illness on the National Occupational Research Agenda (NORA) [NIOSH, 2003]. Gaps remain however, notably: there is much less research on the specific physiological pathways of disease causation although failures in blood pressure regulation, when assessed by ambulatory monitoring, are clearly identified [Belkic et al., 2004]. We have identified regulatory limitations of central nervous system's control (sympathetic and parasympathetic autonomic) of cardiac output as a pathway by which psychosocial risks cause illness. These ANS limitations, if confirmed, could implicate work stress in the broad range of other diseases affected by ANS regulation. If the implicated pathway (the central, basal regulatory power of the ANS, with its sympathetic and parasympathetic mechanisms) is confirmed with further studies, it implicates a broad range of the chronic diseases as potentially associated with job strain. Disruptions in ANS control make a case for a possibility that musculoskeletal, mental strain, or diabetes outcomes could be related to the same basic job-related strains [van der Doef and Maes, 1999; Bongers et al., 2002].

However, further studies are needed to explicitly demonstrate the full sequence of process by which defective cardiac regulation measured by HRV serves as the final CVD disease pathway for job stress.

ACKNOWLEDGMENTS

The authors thank all study participants for their time and commitment to the protocol. The study would not have been possible without the collaborative support of Dr. Ira Ockene, Dr. Robert Mittleman, Karen Rofino-Nadworny, and Philip Merriam (University of Massachusetts Medical Center); Dr. Thomas Bigger and Richard Steinman (Columbia University); Dr. Ronald Berger and Barry Fetics (John Hopkins Medical Center).

REFERENCES

Albrecht P, Cohen RJ. 1988. Estimation of heart rate power spectrum bands from real world data: Dealing with ectopic beats and noisy data. Comp Cardiol 15:311–314.

Algra A. 1993. Heart rate variability form 24-hour holter ECG and the 2 year risk for sudden death. Circulation 88:180–185.

Appels A, Mulder P. 1988. Excess fatigue as a precursor of myocardial infarction. Eur Heart J 9:758–764.

Appels A, Hoppener P, Mulder P. 1988. A questionnaire to assess premonitory symptoms of myocardial infarction. Int J Cardiol 17: 15–24.

Atiga WL, Calkins H, Lawrence JH, Tomaselli GF, Smith JM, Berger RD. 1998. Beat to beat repolarization lability identifies patients at risk for sudden cardiac death. J Cardiovasc Electrophysiol 9:899–908.

Babuty D, Lab M. 2001. Mechanoelectrical contributions to sudden cardiac death. Cardiovasc Res 50:270–279.

Backs RW. 1995. Going beyond heart rate: Autonomic space and cardiovascular assessment of mental workload. Int J Aviation Psychol 5:25–48.

Backs RW. 1998. A comparison of factor analytic methods of obtaining cardiovascular autonomic components for the assessment of mental workload. Ergonomics 41:733–745.

Backs RW, Lenneman J, Sicard JL. 1999. The use of autonomic components to improve cardiovascular assessment of mental workload in flight simulation. Int J Aviation Psychol 9:33–47.

Belkic K, Landsbergis P, Schnall P, Baker D. 2004. Is job strain a major source of cardiovascular disease risk? Scand J Work Environ Health 3:85–128.

Berger RD, Kasper EK, Baughman KL, Marban E, Calkins H, Tomaselli GF. 1997. Beat-to-beat QT interval variability: Novel evidence for repolarization lability in ischemic and nonischemic dilated cardiomyopathy. Circulation 96:1557–1565.

Berntson GG, Bigger JT, Jr., Eckberg D, Grossman P, Kauffman P, Malik M, Nagaraja H, Porges S, Saul JP, Stone PH, Van Der Molen MW. 1997. Heart rate variability: Origins, methods, and interpretative caveats. Psychophysiology 34:623–648.

Bigger JT. 1992. Frequency domain measures of heart period variability and mortality after myocardial infarction. Circulation 85: 164–171.

Bigger JT, Jr. 1994. Spectral analysis of RR variability to evaluate autonomic physiology and pharmacology and to predict cardiovascular outcomes in humans. In: Zipes DP, editor. Cardiac Arrhythmias: From Cell to Bedside. Philadelphia, PA: WB Saunders. p 1151–1170.

Bigger JT, La Rovere RT, Marcus FI, Mortara A, Schwartz PJ. 1998. Baroreflex sensitivity and heart rate variability in prediction of total cardiac mortality after myocardial infarction. Lancet 351:478–484.

Bongers P, Kremer A, Laak J. 2002. Are psychosocial factors, risk factors for symptoms and signs of the shoulder, elbow, or hand/wrist?: A review of the epidemiological literature. Am J Ind Med 41:315–342.

Bultmann U, Kant IJ, Schroer CA, Kasl SV. 2002. The relationship between psychosocial work characteristics and fatigue and psychological distress. Int Arch Occup Environ Health 75:259–266.

Bultmann U, Kant IJ, Van den Brandt PA, Kasl SV. 2002. Psychosocial work characteristics as risk factors for the onset of fatigue and psychological distress: Prospective results from the Maastricht cohort study. Psychol Med 32:333–345.

Cannon WB. 1914. The emergency function of the adrenal medulla in pain and the major emotions. Am J Physiol 33:356–372.

Collins SM. 2001. Emerging methods for the physiological assessment of occupational stress. Work 17:209–219.

Collins SM. 2003. Job strain and electrocardiographically assessed pathophysiological mechanisms. University of Massachusetts Lowell, Sc.D. Dissertation.

Conway J, Boon N, Jones JV, et al. 1984. Neural and humoral mechanisms involved in blood pressure variability. J Hypertens 2:203–208.

Dekker JM. 1997. Heart rate variability from short electrocardiographic recordings predicts mortality from all causes in middle aged and elderly men. Am J Epidemiol 145:899–908.

Doulalas MD, Flather D, Pipilis A, Campbell S. 2001. Evolutionary pattern and prognositic importance of heart rate variability during the early phase of acute myocardial infarction. Int J Cardiol 77:169–179.

Fetsch T, Reinhart L, Wichter T, Borggrefe M, Breithardt G. 1998. Heart rate variability and electrical stability. Basic Res Cardiol 93:S117–S124.

Galinier M, Pathak A, Fourcade J, Andodias C, Curnier D, Varnous S, Boveda S, Massabuau P, Fauvel M, Senard JM, Bounhoure JP. 2000. Depressed low frequency power of heart rate variability as an independent predictor of sudden death in chronic heart failure. Eur Heart J 21:475–482.

Gronefeld G, Hohnloser S. 2001. What do implantable cardioverter/defibrillators teach us about the mechanisms of sudden cardiac death. Cardiovasc Res 50:232–241.

Grossman P, Sveback S. 1987. Respiratory sinus arrythmia as an index of parasympathetic cardiac control during active coping. Psychophysiology 24:228–235.

Guyton AC, Hall JE. 2000. Textbook of medical physiology. 10th edn. Philadelphia: Saunders W B Co.

Jacob RG, Thayer JF, Manuck SB, Muldoon MF, Tamres LK, Williams DM, Ding Y, Gatsonis C. 1999. Ambulatory blood pressure responses and the circumplex model of mood: A 4-day study. Psychosom Med 61:319–333.

Johnson JV, Hall EM. 1988. Job strain, workplace social support, and cardiovascular disease: A crosssectional study of a random sample of the Swedish working population. Am J Pub Health 86:324–331.

Kalsbeek JWH. 1973. Do you believe in sinus arrhythmia? Ergonomics 16:99–104.

Kamarack TW, Shiffman S, Smithline L, et al. 1998. Effects of task strain, social conflict, and emotional activation on ambulatory cardiovascular activity: Daily life consequences of recurring stress in a multiethnic adult sample. Health Psychol 17:17–29.

Karasek RA. 1976. The impact of the work environment on life outside the job. Ph.D. Dissertation: Massachusetts Institute of Technology.

Karasek RA. 1979. Job demands, job decision latitude and mental strain: Implications for job redesign. Adm Sci Q 24:285–308.

Karasek RA. 2005. The stress disequilibrium theory of chronic disease development. International Conference of Occupational Health: Work Environment and Cardiovascular Disease. Los Angeles, CA. March 9–11, 2005.

Karasek RA, Theorell T. 1990. Healthy work: Stress productivity, and the reconstruction of working life. New York: Basic Books.

Karasek RA, Gordon G, Pietroskovsky C. 1985. Job content instrument: Questionnaire and user's guide. Los Angelas, CA/Lowell, MA: University of Southern California/University of Massachusetts.

Karasek RA, Brisson C, Kawakami N. 1998. The job content questionnaire: An instrument for internationally comparative assessment of psychosocial job characteristics. J Occup Health Psychol 3:322–355.

Kleiger RE, Miller JP, Bigger JT, Jr., Moss AJ. 1987. Decreased heart rate variability and its association with increased mortality after acute myocardial infarction. Am J Cardiol 89:256–262.

Kobayashi F, Akamatsu Y, Furui H, Watanabe T, Horibe H. 1997. Effects of night shift on psychophysiological parameters among hospital nurses—influence of changing from a full-day to half-day work shift before night duty. Int Arch Occup Environ Health 69:83–90.

KOSHA. 1998. Annual report inchon. Korea: Korean Occupational Health and Safety Agency.

Kristal-Boneh E, Raifal M, Froom P, Ribak J. 1995. Heart rate variability in health and disease. Scand J Work Environ Health 21:85–95.

Kristenson TS. 1995. The demand-control-support model: Methodological challenges and future research. Stress Med 11:17–26.

Kristenson TS. 1996. Job stress and CVD: A theoretic critical review. J Occup Health Psychol 1:246–260.

Kuch B, Hense HW, Sinnreich R, Kark JD, von Eckardstein A, Sapoznikov D, Bolte HD. 2000. Determinants of short period heart rate variability in the general population. Cardiology 95:131–138.

Landsbergis P, Theorell T. 2000. Measurement of psychosocial workplace exposure variables: Self-report questionnaires. In: Schnall P, Belkic K, Landsbergis P, Baker D, editors. The workplace and cardiovascular disease. Philadelphia: Hanley & Belfus.

Lenneman J, Backs RW. 2000. The validity of factor analytically derived cardiac autonomic components for mental workload assessment. In: Backs RW, Boucsein W, editors. Engineering psychophysiology: Issues and applications, London: Lawrence Erlbaum. pp 161–176.

Lombardi F, Sandrone F, Spinnler M, Torzilla D, Lavezzaro G, Brusca D, Malliani A. 1996. Heart rate variability in the early hours of an acute myocardial infarction. Am J Cardiol 77:1037–1044.

Lombardi F, Makikallio TH, Myerburg R, Huikuiri H. 2001. Sudden cardiac death: Role of heart rate variability to identify patients at risk. Cardiovasc Res 50:210–217.

Lucini D, Mela GS, Malliani A, Pagani M. 2002. Impairment in cardiac autonomic regulation preceding arterial hypertension in humans: Insights from spectral analysis of beat-by-beat cardiovascular variability. Circulation 106:2673–2679.

Magari SR, Hauser R, Schwartz J, Williams PL, Smith TJ, Christiani DC. 2001. Association of heart rate variability with occupational and environmental exposure to particulate air pollution. Circulation 14:986–991.

Magari SR, Schwartz J, Williams PL, Hauser R, Smith TJ, Christiani DC. 2002. The association of particulate air metal concentrations with heart rate variability. Environ Health Perspect 110:875–880.

Malliani A, Pagani M, Lombardi F, Cerutti S. 1991. Cardiovascular neural regulation explored in the frequency domain. Circulation 84: 482–492.

Merriam PA, Ockene IS, Hebert JR, Rosal MC, Matthews CE. 1999. Seasonal variations of blood cholesterol levels study: Study methodology. J Biol Rhythms 14:330–339.

Mortara A, LaRovere MT. 1994. Can power spectral analysis of heart rate variability identify a subgroup of high risk patients with CHF with sympathetic activation. Br Heart J 71:422–430.

Myerburg R, Spooner P. 2001. Opportunities for sudden cardiac death prevention: Directions for new clinical and basic research. Cardiovasc Res 50:177–185.

Negoescu R, Dinca-Panaitescu S. 1997. Mental stress enhances the sympathetic fraction of QT variability in an RR independent way. Int Phys Behav Sci 32:220–226.

NIOSH. 2003. National Occupational Research Agenda: Update 2003. DHHS (NIOSH) Publication Number 2003-148.

Nolan J. 1998. Prospective study of heart rate variability and mortality in chronic heart failure. Circulation 98:1510–1516.

Pagani M, Rimordi O, Pizzinelli R. 1991a. Assessment of the neural control of the circulation during psychological stress. J Autonomic Nervous Syst 35:33–42.

Pagani M, Mazzuero G, Ferrari A. 1991b. Sympathovagal interaction during mental stress: A study using spectral analysis of heart rate variability in healthy control subjects and patients with a prior history of myocardial infarction. Circulation 83(Suppl II):II43–II51.

Phillips AN, Smith GD. 1993. The design of prospective epidemiological studies: More subjects or better measurements? J Clin Epidemiol 46:1203–1211.

Prescott E, Holst C, Grønbæk M, Schnohr P, Barefoot J. 2003. Vital exhaustion as a risk factor for ischaemic heart disease and all cause mortality in a community sample. A prospective study of 4084 men and 5479 women in the Copenhagen City Heart Study. Int J Epidemiol 32:990–997.

Rottman J, Steinman RC, Albrecht P, Bigger JT, Jr., Rolnitzky LM, Fleiss JL. 1990. Efficient estimation of the heart period power spectrum suitable for physiologic or pharmacologic studies. Am J Cardiol 66: 1522–1524.

Schnall P, Pieper C, Schwartz JE, Karasek RA, et al. 1990. The relationship between ''Job Strain,'' workplace diastolic blood pressure and left ventricular mass index. JAMA 263:1929–1935.

Schnall P, Landsbergis P, Baker D. 1994. Job strain and CVD. Annu Rev Public Health 15:381–411.

Schnall P, Schwartz JE, Landsbergis P, Warren K, Pickering TG. 1998. A longitudinal study of job strain and ambulatory blood pressure: Results from a three year follow-up. Psychosom Med 60:697–706.

Schnall PL, Belkic K, Landsbergis P, Baker D, editors. 2000. The workplace and cardiovascular disease. Philadelphia: Hanley & Belfus, INC.

Schwartz J, Pickering T. 1996. The Workplace Blood Pressure Study. American Society of Hypertesion Conference.

Siegrist J, Junge PR, Cremer P, Seidel D. 1990. Low status control, high effort at work and ischemic heart disease: Prospective evidence from blue collar men. Soc Sci Med 31:1127–1134.

Singh JP, Larson MG, Tsuji H, Evans JC, O'Donnell CJ, Levy D. 1998. Reduced heart rate variability and new onset hypertension. Hypertension 32:293–297.

Sloan RP, Shapiro PA, Bagiella E, Boni S, Paik M, Bigger JT, Jr., Steinman RC, Gorman JM. 1994. Effect of mental stress throughout the day on cardiac autonomic control. Biol Psychol 37:89–99.

Stein PK, Kleiger RE. 1999. Insights from the study of heart rate variability. Ann Rev Medicine 50:249–261.

Steptoe A. 2000. Stress, social support and cardiovascular activity over the working day. Int J Psychophysiol 37:299–308.

Steptoe A, Fieldman G, Evans O, Perry L. 1996. Cardiovascular risk and responsivity to mental stress: The influence of age, gender, and risk factors. J Cardiovasc Risk 3:83–93.

Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. 1996. Heart rate variability standards of measurement, physiological interpretation, and clinical use. Eur Heart J 17:354–381.

Theorell T, Perski A, Orth-Gomér K, Hamsten A, de Faire U. 1991. The effects of the strain of returning to work on the risk of cardiac death after first myocardial infarction before the age of 45. Int J Cardiol 30:61–76.

Tusji H, Larson MG. 1996. Impact of reduced HRV on risk for cardiac events: The Framingham Heart Study. Circulation 94:2850–2855.

Tusji H, Vendetti FJ. 1994. Reduced HRV and mortality risk in an elderly cohort: The Framingham Heart Study. Circulation 90:878–883.

Tzaneva L, Danev S, Nikolova R. 2001. Investigation of noise exposure effect on heart rate variability. Central Eur J Pub Health 9:130–132.

van Amelsvoort LGPM, Schouten EG, Maan AC, Swene CA, Kok FJ. 2000. Occupational determinants of heart rate variability. Int Arch Occup Environ Health 73:255–262.

van Boven AJ, Crijns H, Haaksma J, Zwinderman AH. 1998. Depressed heart rate variability is associated with events in patients with stable coronary artery disease and preserved left ventricular dysfunction. Am Heart J 135:571–576.

Van der Doef M, Maes S. 1999. The job demand-control (-support) model and psychological well-being: A review of 20 years of empirical research. Work Stress 13:87–114.

Vrijkotte TGM, van Dornen LPJ, de Geus EJC. 2000. Effects of work stress on ambulatory blood pressure, heart rate, and heart rate variability. Hypertension 35:880–886.

Wantanabe T, Sugiyama Y, Sumi Y, Wantanabe M, Takeuchi K, Kobyashi F, Kono K. 2002. Effects of vital exhaustion on cardiac autonomic nervous functions assessed by heart rate variability at rest in middle-aged male workers. Int J Behav Med 9:68–75.

Yeragani VK, Pohl R, Jampala VC, Balon R, Kay J, Igel G. 2000. Effect of posture and isoproterenol on beat-to-beat heart rate and QT variability. Neuropsychobiology 41:113–123.

Yeragani VK, Pohl R, Jampala VC, Balon R, Ramesh C, Srinivasan K. 2000. Increased QT variability in patients with panic disorder and depression. Psychiatry Res 93:225–235.

Appendix: Triangulation of Job Strain Measures

The triangulation weights were based on the working hypothesis that the demand–control model is an effective predictor of emotion and that thus the best weighting formula would be attained by assessing which methods display the strongest association with emotion in a multiple response relationship [Karasek and Theorell, 1990; Van der Doef

and Maes, 1999; Bultmann et al., 2002] using the emotional state responses to questions on the JCQ (4) and diary (6)). Demands and control scores from each method were entered into linear regression models to predict: (1) negative emotion (seven items, Cronbach's Alpha 0.77); and (2) positive emotion (three items, Cronbach's Alpha 0.66). Coefficients for occupational code demands were not significantly associated with emotion and were not included in weighting. Standardized regression coefficients for the job strain assessment methods were summed and the percent of the sum accounted for by each method was utilized to weight the scores. Seventy percent of the total job strain score was determined by negative emotion (seven questions) and 30% were determined based on the positive emotion (three questions). The correlation structure between measures for demands and control are presented in Figure A1, and show substantial consistency across all three methods for decision latitude, and in the case of demands: for diary and questionnaire.

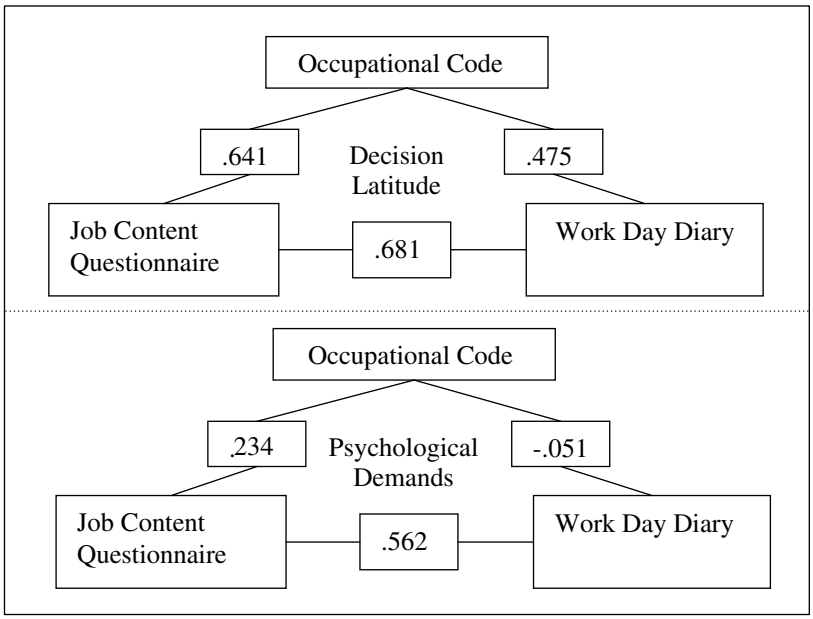


FIGURE A1. Triangulation of three job strain measures.